Krishna Karthikeyan | BE23B033 | IIT Madras | KTH Stockholm, Sweden| Computational Modeling & AI

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
Bachelor of Technology in Biological Engineering	Indian Institute of Technology, Madras	8.87	2027
Exchange Semester	KTH Royal Institute of Technology	Completed	2025-26
Senior Secondary Class XII (CBSE)	Chettinad Vidyashram	94.8%	2023
Secondary Class X (CBSE)	Chettinad Vidyashram	92.4%	2021

- Awarded the **KVPY Fellowship** by DST, Government of India, with **All India Rank 48**.
- Secured **All India Rank 6025** in JEE Advanced, **99.2 percentile** in JEE Main and **1560/1600 in the SAT**.
- Completed the **one-year Foundation Level of the IIT Madras BS in Data Science and Applications**.
- Selected for an exchange semester at **KTH Royal Institute of Technology, Stockholm**, completing 81 transfer credits.

RELEVANT COURSES & SKILLS

Relevant Courses	**Computational Systems Biology; Bioinformatics; Analysis and Interpretation of Biological Data; Fundamentals of Deep Learning; Differential Equations**; Signals and Systems; Biochemical Thermodynamics; Numerical Methods in Biology; Cellular Engineering*; Molecular Basis of Diseases*; **Recent Advances in Generative AI***
Skills	Mathematical and computational modeling; systems biology; biological-network analysis; statistical modeling; machine and deep learning; scientific computing; numerical optimization; biomedical-data integration and visualization; model evaluation; generative-AI workflow development; research communication
Software	Python, R, MATLAB, Bash/Linux, PyTorch, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, FastAPI, OpenSim, VTK, PyVista, PyQt5, Git and Jupyter Notebook

PROFESSIONAL EXPERIENCE

GenePath Diagnostics Pune
May '26 - Jun '26

Bioinformatics Intern | Metagenomics Pipeline

- Built a reproducible end-to-end workflow converting raw **16S rRNA and ITS-1 FASTQ data** into quality-controlled amplicon sequence variants and taxonomic profiles.
- Implemented **DADA2 filtering, dereplication, denoising, chimera removal and paired-end merging** and integrated SILVA, RDP and Greengenes2 reference databases.
- Developed a locally deployable **FastAPI application** for FASTA, FASTQ and POD5 inputs, optional QIIME2 processing, data-quality summaries, visualizations and downloadable outputs; validated it on water-sample data.
- Observed clinical molecular-diagnostics workflows including DNA/RNA extraction, PCR, BioFire and KingFisher platforms; contributed to a PCR Interpreter tool.

Repository: github.com/DragonEolf/Amplicon2TaxonomyPipeline

RESEARCH AND COMPUTATIONAL PROJECTS

Epistasis-Function Systems Biology Research*
Aug '26 - present

[Ongoing Research - Computational Systems Biology, IIT Madras]

- Investigating whether **genome-scale epistatic interactions predict bacterial colonization and persistence in the mouse gut**, extending prior computational work on lactic acid bacteria under Prof. Karthik Raman and Prof. Meiyappan Lakshmanan.
- Developing a computational framework to relate **epistatic-network characteristics across bacterial strains to in vivo fitness and survival**, testing the hypothesis that greater epistatic connectivity is associated with enhanced ecological adaptability.

Deep Learning Neural Networks
Jan '26 - Apr '26

Fundamentals of Deep Learning | CS6910, IIT Madras

- Implemented and evaluated **deep neural networks for image classification**, including feed-forward networks, CNNs and stacked autoencoders, with systematic analysis of test-set performance and class-wise errors.
- Experimented with **SGD, Momentum, AdaGrad, RMSProp and Adam**, alongside Batch and Layer Normalization, analyzing their effects on convergence, generalization and classification performance.
- Built a **stacked-autoencoder pretraining pipeline** for representation learning from unlabeled data and evaluated learned latent representations against end-to-end supervised deep networks.

Medical Data Visualization
Oct '25 - Jan '26

[Course Project - Medical Image Visualization, KTH]

- Built a Python tool using **VTK, PyVista and PyQt5** for CT, CTA and MRA volume rendering, surface reconstruction and interactive visualization of segmented cerebral vasculature.
- Implemented anatomical labeling, artifact-aware preprocessing and repeatable screenshot and fly-through exports.

Biomechanical Model & Simulation
Oct '25 - Jan '26

[Course Project - Biomechanics of Human Movement, KTH]

- Used **OpenSim and MATLAB** inverse kinematics, inverse dynamics and static optimization to model gait and evaluate muscle activation and metabolic-cost effects of spring parameters.

Generative-AI Translation Studio*
Aug '26 - present

[Ongoing Course Project - Recent Advances in Generative AI]

- Building an end-to-end **generative-AI translation studio**, integrating a model-assisted translation workflow into a working user-facing application and codebase.
- Structuring the workflow for reproducible implementation and demonstrable outputs.

3D Knapsack Optimization
Nov '24 - Dec '24

[Inter-IIT Competition Project]

- Placed **4th nationwide among 23 IITs** by developing a scalable package-allocation strategy under weight, delivery-priority and vertical-stability constraints.
- Designed Neighbor Aligned Cornering and Layering algorithms to improve space utilization and reduce Unit Load Device usage.

TEACHING AND SCIENTIFIC COMMUNICATION

Teaching & Scientific Outreach
2025 - present

- Supporting Engineering Mechanics students with quantitative problem solving while coordinating with the instructional team.
- Delivered a **six-hour biotechnology workshop for 30 participants** through interactive instruction and hands-on gel electrophoresis.

***Ongoing course/project**